

Math 115A Final

TOTAL POINTS

98 / 100

QUESTION 1

Characteristic polynomials and eigenvalues 20 pts

1.1 Distinct eigenvalues 10 / 10

✓ - 0 pts Correct

- 2 pts incorrect matrix representation
- 1 pts incorrect characteristic polynomial
- 1 pts incorrect eigenvalues

1.2 Repeated eigenvalues 10 / 10

✓ - 0 pts Correct

- 2 pts incorrect matrix representation
- 1 pts incorrect characteristic polynomial
- 1 pts incorrect eigenvalues

QUESTION 2

20 pts

2.1 Difference between S and $\text{span}(S)$ 10 / 10

✓ - 0 pts Correct

- 7 pts incorrect but tried something
- 5 pts did not provide a correct counterexample
- 1 pts miscellaneous mistake

2.2 linear (in)dependence 8 / 10

- 0 pts Correct
- 7 pts incorrect but tried something

- 5 pts did not provide a correct counterexample

- 1 pts miscellaneous mistake

- 2 Point adjustment

💬 proved the statement for $R_2[x]$, not R^3

QUESTION 3

3 An intersection between two eigenspaces 10 / 10

✓ - 0 pts Correct

- 4 pts incorrect setup: did not define $\vec{v} \in E_{\lambda_1}(T) \cap E_{\lambda_2}(T)$

- 4 pts did not apply the cancellation law to show $\vec{v} = \vec{0}$.

- 1 pts miscellaneous mistake

- 5 pts Using incorrect assumptions

- 3 pts incorrect setup

QUESTION 4

Cauchy-Schwartz inequality 20 pts

4.1 Cauchy-Schwartz inequality on the standard inner product space 10 / 10

✓ - 0 pts Correct

- 5 pts incorrect usage of the inner product

- 1 pts miscellaneous mistake

4.2 Application of Cauchy-Schwartz inequality 10 / 10

✓ - 0 pts Correct

- 8 pts incorrect reasoning but tried something
- 1 pts computational mistake
- 5 pts incorrect application of 4.(a)
- 9 pts incorrect

QUESTION 5

5 The matrix representation of a linear

map 10 / 10

✓ - 0 pts Correct

- 0.5 pts not finding $T(\frac{1}{2}, \frac{\sqrt{3}}{2})$
- 0.5 pts not finding $T(-\frac{\sqrt{3}}{2}, \frac{1}{2})$
- 4 pts not finding $T(\frac{1}{2}, \frac{\sqrt{3}}{2})_B$
- 4 pts not finding $T(-\frac{\sqrt{3}}{2}, \frac{1}{2})_B$
- 2 pts incorrect $T(\frac{1}{2}, \frac{\sqrt{3}}{2})_B$
- 2 pts incorrect $T(-\frac{\sqrt{3}}{2}, \frac{1}{2})_B$
- 9 pts No reasoning

QUESTION 6

Adjoint and orthogonal complements 20 pts

6.1 10 / 10

✓ - 0 pts correct

- 4 pts incorrect proof for $(\mathrm{im} T^\perp)^\perp \subset \ker T$
- 4 pts incorrect proof for $\ker T \subset (\mathrm{im} T^\perp)^\perp$

- 1 pts Miscellaneous mistakes

- 2 pts incorrect proof for

$(\mathrm{im} T^\perp)^\perp \subset \ker T$ but the direction was correct

- 9 pts incorrect

- 2 pts incorrect proof for $\ker T \subset$

$(\mathrm{im} T^\perp)^\perp$ but the direction was correct

6.2 10 / 10

✓ - 0 pts Correct

- 4 pts applying theorems about dimensions in an incorrect way
- 7 pts tried to show $\ker T = \ker T^*$
- 8 pts incorrect reasoning
- 10 pts incorrect

Math 115A
Linear Algebra
Final

Question	Points	Score
1	20	
2	20	
3	10	
4	20	
5	10	
6	20	
Total:	100	

Some definitions: (V and W are vector spaces over $\mathbb{R}$)

- $\text{span}(S)$: the set of all linear combinations of the vectors in S
- $B \subset V$ is a basis of V if $\text{span}(B) = V$ and B is linearly independent
- $T: V \rightarrow W$ is an isomorphism if T is an invertible linear map.
- If $T(\vec{v}) = \lambda \vec{v}$ and $\vec{v} \neq \vec{0}$, then $\vec{v}$ is an eigenvector of T with eigenvalue λ .
- $E_\lambda(T) := \ker(T - \lambda I)$. ($I: V \rightarrow V$ is the identity map)
- The characteristic polynomial of T . $p(\lambda) := \det([T]_B - \lambda I_n)$. (I_n is the identity matrix)
- The standard inner product space $(\mathbb{R}^2, \langle, \rangle)$: $\langle (a, b), (x, y) \rangle = ax + by$.
- $\|\vec{v}\| := \sqrt{\langle \vec{v}, \vec{v} \rangle}$
- $B = \{\vec{u}_1, \dots, \vec{u}_n\}$ is an orthonormal basis if B is a basis and $\|\vec{u}_i\| = 1$ for $1 \leq i \leq n$ and $\langle \vec{u}_i, \vec{u}_j \rangle = 0$ for $i \neq j$.
- The orthogonal complement of W . $W^\perp := \{\vec{v} \in V : \langle \vec{v}, \vec{w} \rangle = 0 \text{ for any } \vec{w} \in W\}$.
- $\langle T(\vec{v}), \vec{w} \rangle = \langle \vec{v}, T^*(\vec{w}) \rangle$ for all $\vec{v}, \vec{w} \in V$.

Some theorems, propositions and corollaries: (V and W are vector spaces over $\mathbb{R}$)

- $\text{span}(S)$ is a subspace of V .
- $T: V \rightarrow W$ is injective if and only if $\ker T = \{\vec{0}\}$.
- $T: V \rightarrow W$ is an isomorphism if T is injective and surjective.
- If $T: V \rightarrow V$ and V is finite dimensional, then T is injective if and only if T is surjective.
- $|\langle \vec{v}, \vec{w} \rangle| \leq \|\vec{v}\| \|\vec{w}\|$. (Cauchy-Schwartz inequality)
- If $B = \{\vec{u}_1, \dots, \vec{u}_n\}$ is an orthonormal basis of V , then $[\vec{v}]_B = \langle \vec{v}, \vec{u}_1 \rangle \vec{u}_1 + \dots + \langle \vec{v}, \vec{u}_n \rangle \vec{u}_n$
- $\vec{v} = \vec{0}$ if and only if $\|\vec{v}\| = 0$.
- If $\langle \vec{v}, \vec{w}_1 \rangle = \langle \vec{v}, \vec{w}_2 \rangle$ for all $\vec{v} \in V$, then $\vec{w}_1 = \vec{w}_2$.
- $\dim V = \dim \ker T + \dim \text{im } T$. (rank-nullity theorem)
- $\dim V = \dim W + \dim W^\perp$.
- $V = W \oplus W^\perp$

1. Let $B = \{(1, 0), (0, 1)\}$ be an ordered basis of $\mathbb{R}^2$. Find the characteristic polynomial and eigenvalues of $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$.

(a) (10 points) $T(x, y) = (15x + y, 11x + 5y)$

$$T(1, 0) = (15, 11) = 15 \cdot (1, 0) + 11 \cdot (0, 1)$$

$$T(0, 1) = (1, 5) = 1 \cdot (1, 0) + 5 \cdot (0, 1)$$

$$[T]_B = \begin{pmatrix} 15 & 1 \\ 11 & 5 \end{pmatrix} \quad \det \begin{pmatrix} 15-\lambda & 1 \\ 11 & 5-\lambda \end{pmatrix} = (15-\lambda)(5-\lambda) - 11$$
$$= \lambda^2 - 20\lambda + 75 - 11$$
$$= \lambda^2 - 20\lambda + 64$$

$$P(\lambda) = (\lambda - 4)(\lambda - 16)$$

$$\lambda = 4, 16$$

eigenvalues are 4 and 16.

(b) (10 points) $T(x, y) = (10x + 3y, -3x + 4y)$

$$T(1, 0) = (10, -3) = 10 \cdot (1, 0) - 3(0, 1)$$

$$T(0, 1) = (3, 4) = 3 \cdot (1, 0) + 4(0, 1)$$

$$[T]_B = \begin{pmatrix} 10 & 3 \\ -3 & 4 \end{pmatrix}$$

$$\det \begin{pmatrix} 10-\lambda & 3 \\ -3 & 4-\lambda \end{pmatrix} = (10-\lambda)(4-\lambda) + 9 = \lambda^2 - 14\lambda + 49 = (\lambda-7)^2$$

$$P(\lambda) = (\lambda-7)^2$$

$$\lambda = 7$$

eigenvalue is 7.

2. If the following statement is true, prove it. Otherwise, provide a counterexample.

(a) (10 points) Let S be a subset of $\mathbb{R}_2[x]$. If $p(x) \in \text{span}(S)$, then $p(x) \in S$.

False. Let $S = \{x\}$, $p(x) = 3x \in \text{span}(S)$

as shown, $p(x) = 3x \in \text{span}(S)$ and $p(x) = 3x \notin S$

- (b) (10 points) Let $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$ be non-zero vectors. Suppose $\vec{v}$ is not a scalar multiple of $\vec{u}$, and $\vec{w}$ is not a scalar multiple of $\vec{u}$. Then $S = \{\vec{v}, \vec{w}\}$ is linearly independent.

False. Let $\vec{v} = x^2$, $\vec{w} = 2x^2$, $\vec{u} = x$

Then $\vec{v}$ is not a scalar multiple of $\vec{u}$,
 $\vec{w}$ is not a scalar multiple of $\vec{u}$.

Let $c_1, c_2 = 0$

$$c_1 \vec{v} + c_2 \vec{w} = 0$$

$$c_1 \cdot x^2 + c_2 \cdot 2x^2 = 0$$

$c_1 = -2c_2$, c_1 and c_2 does not have to be 0.

$\vec{v}$ and $\vec{w}$ are scalar multiples of each other

$\Rightarrow S = \{\vec{v}, \vec{w}\}$ is linearly dependent

3. (10 points) Let V be a finite dimensional vector space over $\mathbb{R}$ and $T: V \rightarrow V$ be a linear map. Suppose λ_1 and λ_2 are two distinct eigenvalues of T . Prove $E_{\lambda_1}(T) \cap E_{\lambda_2}(T) = \{\vec{0}\}$.

Take $\vec{v} \neq \vec{0}$; $\vec{v} \in E_{\lambda_1}(T) \cap E_{\lambda_2}(T)$

$$\vec{v} \in E_{\lambda_1}(T) \Rightarrow T(\vec{v}) = \lambda_1 \vec{v}$$

$$\vec{v} \in E_{\lambda_2}(T) \Rightarrow T(\vec{v}) = \lambda_2 \vec{v}$$

$$\Rightarrow T(\vec{v}) = \lambda_1 \vec{v} = \lambda_2 \vec{v}$$

$$\Rightarrow \lambda_1 = \lambda_2$$

contradicts λ_1 and λ_2 are distinct, $\vec{v} = \vec{0}$

$$\text{Let } \vec{v} = \vec{0}, \text{ then } T(\vec{v}) = T(\vec{0}) = \vec{0} = \lambda_1 \vec{v} = \lambda_2 \vec{v}$$

$$\therefore \vec{0} \in E_{\lambda_1}(T) \cap E_{\lambda_2}(T)$$

$$\therefore E_{\lambda_1}(T) \cap E_{\lambda_2}(T) = \{\vec{0}\}$$

4. Let $(\mathbb{R}^2, \langle, \rangle)$ be the standard inner product space.

(a) (10 points) For any $a, b, x, y \in \mathbb{R}$, prove $(ax + by)^2 \leq (a^2 + b^2)(x^2 + y^2)$.

$$|\langle (a, b), (x, y) \rangle| \leq \|(a, b)\| \cdot \|(x, y)\|$$

$$|(ax + by)| \leq \sqrt{a^2 + b^2} \sqrt{x^2 + y^2} = \sqrt{(a^2 + b^2)(x^2 + y^2)}$$

$$\therefore (ax + by)^2 \leq (a^2 + b^2)(x^2 + y^2)$$

- (b) (10 points) Suppose $x^2 + y^2 = 5$. What is the maximum value of $x + 2y$?
(hint: use 4.(a))

$$\text{let } a=1, \quad b=2$$
$$x+2y = \langle (1, 2), (x, y) \rangle$$

$$\text{by 4.(a), } (x+2y)^2 \leq (1^2 + 2^2)(x^2 + y^2) = 5 \cdot 5 = 25$$

$$x+2y \leq 5$$

The maximum of $x+2y$ is 5

5. (10 points) Let $(\mathbb{R}^2, \langle, \rangle)$ be the standard inner product space and $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear map defined by

$$T(x, y) = (x, 0)$$

Let $B = \{(\frac{1}{2}, \frac{\sqrt{3}}{2}), (-\frac{\sqrt{3}}{2}, \frac{1}{2})\}$ be an orthonormal basis. Find $[T]_B$.

$$\begin{aligned} T\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) &= \left(\frac{1}{2}, 0\right) & \left\langle \left(\frac{1}{2}, 0\right), \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \right\rangle &= \frac{1}{4} \\ & & \left\langle \left(\frac{1}{2}, 0\right), \left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right) \right\rangle &= -\frac{\sqrt{3}}{4} \end{aligned}$$

$$\begin{aligned} T\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right) &= \left(-\frac{\sqrt{3}}{2}, 0\right) & \left\langle \left(-\frac{\sqrt{3}}{2}, 0\right), \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \right\rangle &= -\frac{\sqrt{3}}{4} \\ & & \left\langle \left(-\frac{\sqrt{3}}{2}, 0\right), \left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right) \right\rangle &= \frac{3}{4} \end{aligned}$$

$$[T]_B = \begin{pmatrix} \frac{1}{4} & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}$$

6. Let $(V, \langle \cdot, \cdot \rangle)$ be a finite dimensional inner product space over $\mathbb{R}$. $T: V \rightarrow V$, linear

(a) (10 points) Prove that $(\text{im } T^*)^\perp = \ker T$.

Let $\vec{v} \in (\text{im } T^*)^\perp$ Then for all $T^*(\vec{u}) \in \text{im } T^*$ ($\vec{u} \in V$)

$$\langle \vec{v}, T^*(\vec{u}) \rangle = 0$$

$$= \langle T(\vec{v}), \vec{u} \rangle$$

$$= \langle \vec{0}, \vec{u} \rangle = 0$$

Since $\vec{u}$ is any $\vec{u} \in V$, $T(\vec{v}) = \vec{0}$

$\therefore \vec{v} \in \ker T$, $(\text{im } T^*)^\perp \subset \ker T$

Let $\vec{v} \in \ker T$, then $T(\vec{v}) = \vec{0}$

Let $\vec{u}$ be any $\vec{u} \in V$

$$\text{then } \langle T(\vec{v}), \vec{u} \rangle = \langle \vec{0}, \vec{u} \rangle = 0$$

$$= \langle \vec{v}, T^*(\vec{u}) \rangle = 0$$

Since $T^*(\vec{u}) \in \text{im } T^*$

$$\vec{v} \in (\text{im } T^*)^\perp$$

$$\therefore \ker T \subset (\text{im } T^*)^\perp$$

$$\therefore \ker T = (\text{im } T^*)^\perp$$

(b) (10 points) Prove $\dim(\ker T) = \dim(\ker T^*)$. (hint: use 6.(a))

$$\text{Since } (\operatorname{im} T^*)^\perp \oplus (\operatorname{im} T^*) = V \quad (\operatorname{im} T^* < V)$$

$$\dim(\operatorname{im} T^*)^\perp + \dim(\operatorname{im} T^*) = \dim V$$

$$\text{by 6.(a), } \dim(\operatorname{im} T^*)^\perp = \dim(\ker T)$$

$$\dim \ker T = \dim V - \dim(\operatorname{im} T^*)$$

$$\text{also, } \dim \operatorname{im} T^* + \dim \ker T^* = \dim V$$

$$\dim \ker T^* = \dim V - \dim(\operatorname{im} T^*) = \dim \ker T$$

$$\Rightarrow \dim(\ker T^*) = \dim(\ker T)$$

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